

Governance

Reasoning

Perception

CORTEX v8.1

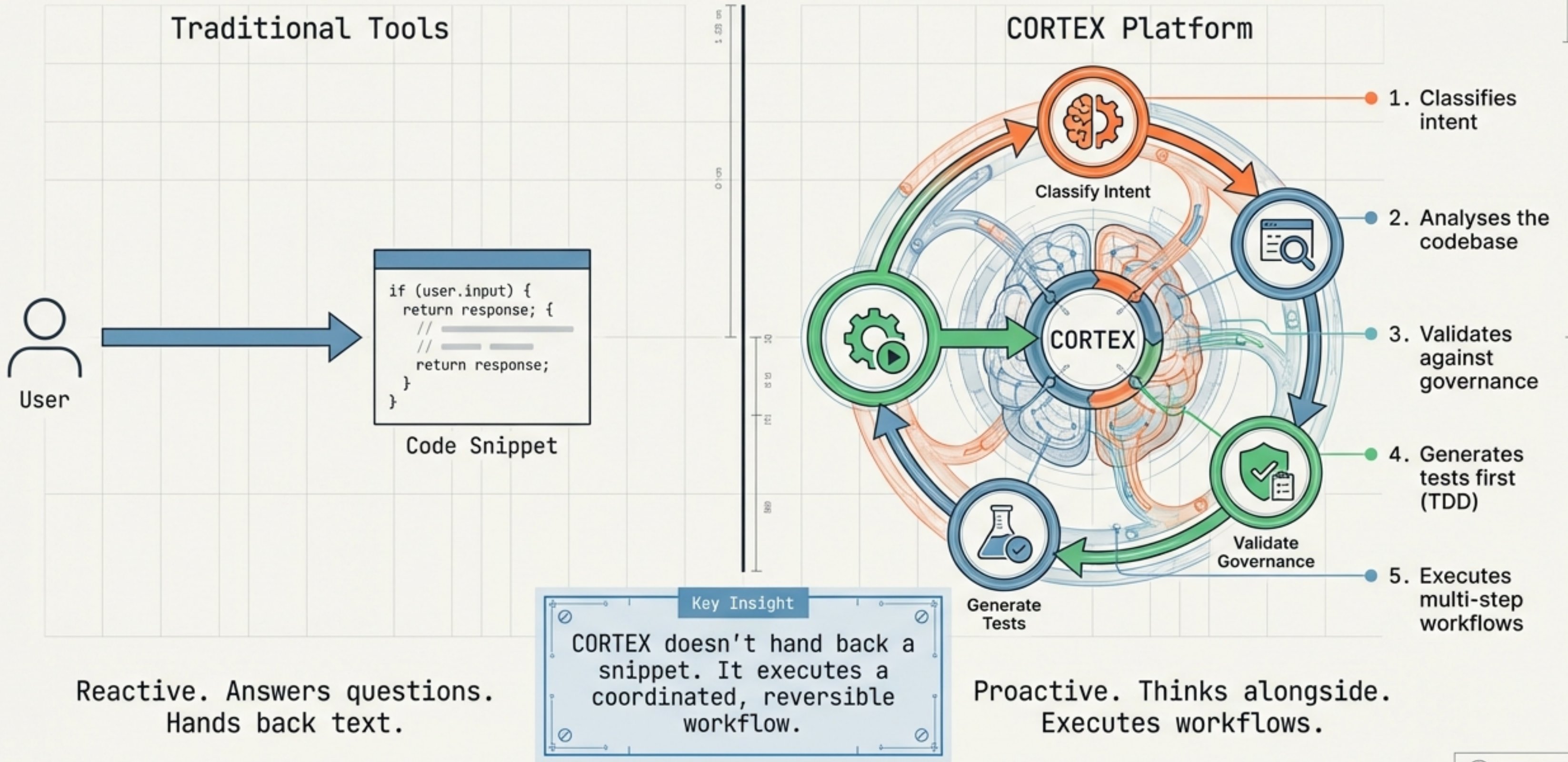
The Intelligent Development Acceleration Platform

Moving beyond code completion to cognitive orchestration.

An AI-native platform combining a cognitive intelligence layer, automated governance, and a rich orchestration engine.

Ship faster. With confidence.

The Core Shift: From Reactive to Cognitive



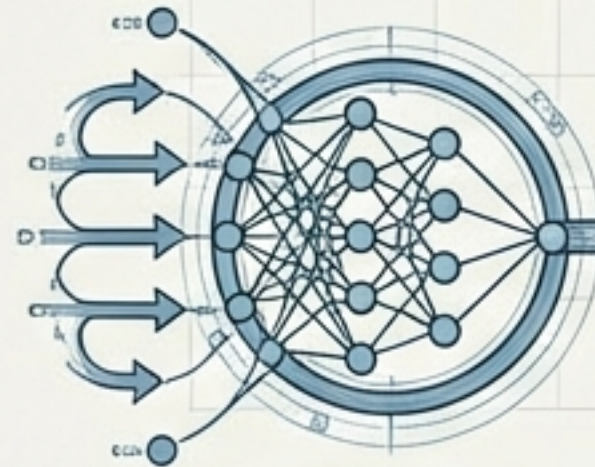
Anatomical Overview: The Six Capability Domains

AI & Intelligence (LENS)

The Eyes

8-analyzer parallel
code understanding

300-800ms
analysis



Decisioning

The Nervous System

Intent routing to 20+ orchestrators

95%+ accuracy

Brain Tier

The Brain

Perception → Reasoning → Action

Confidence
Scored 0.0-1.0



Governance

The Conscience

4-layer enforcement
of 59 CORE rules

<150ms
validation

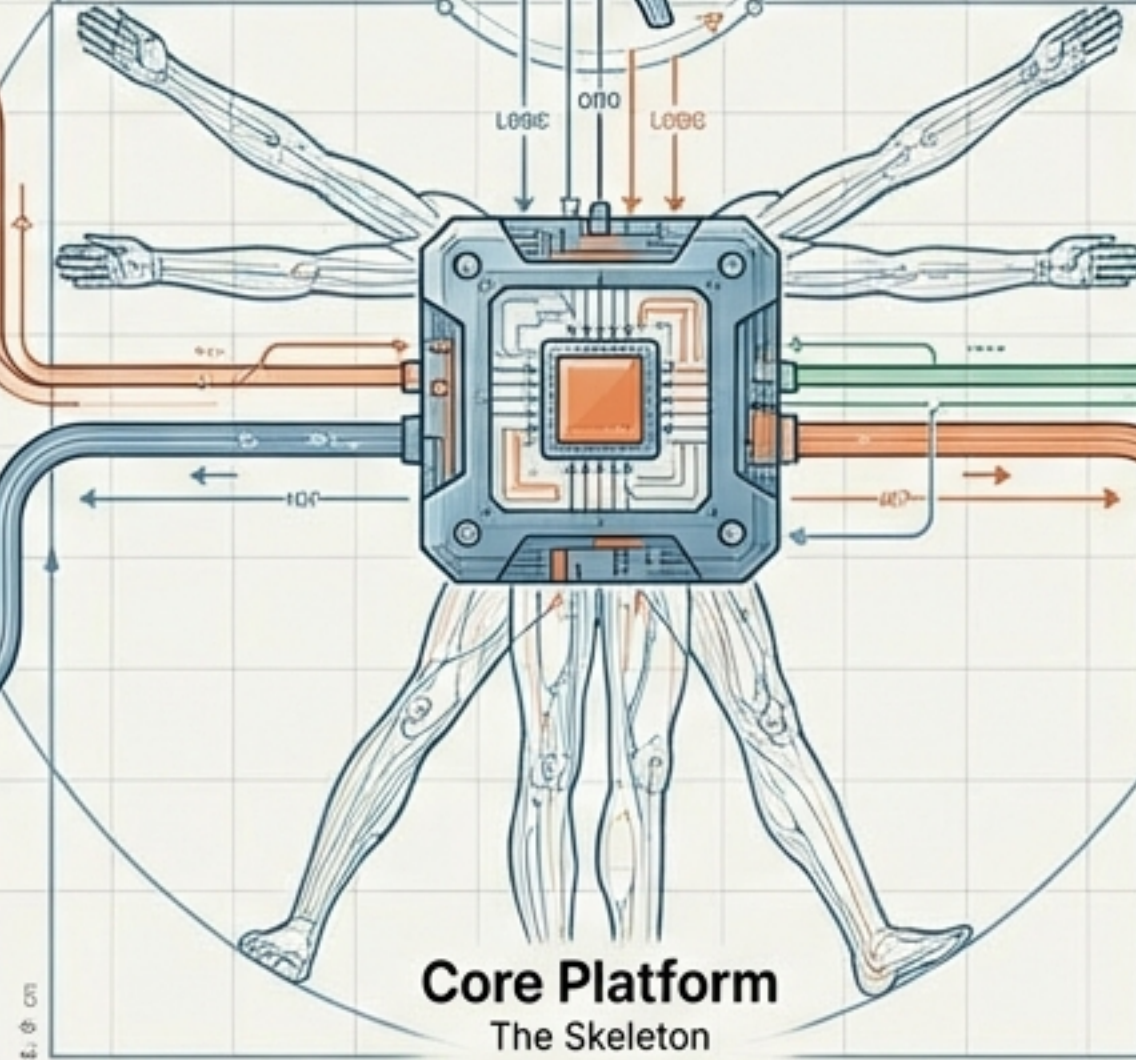


Core Platform

The Skeleton

MCP gateway & orchestration

P50 latency 5ms

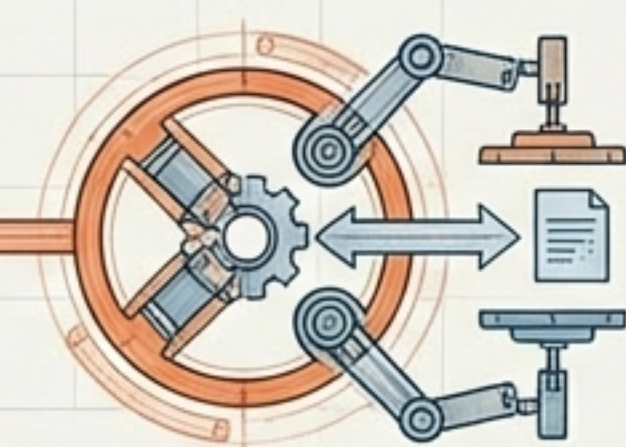


Extensibility

The Hands

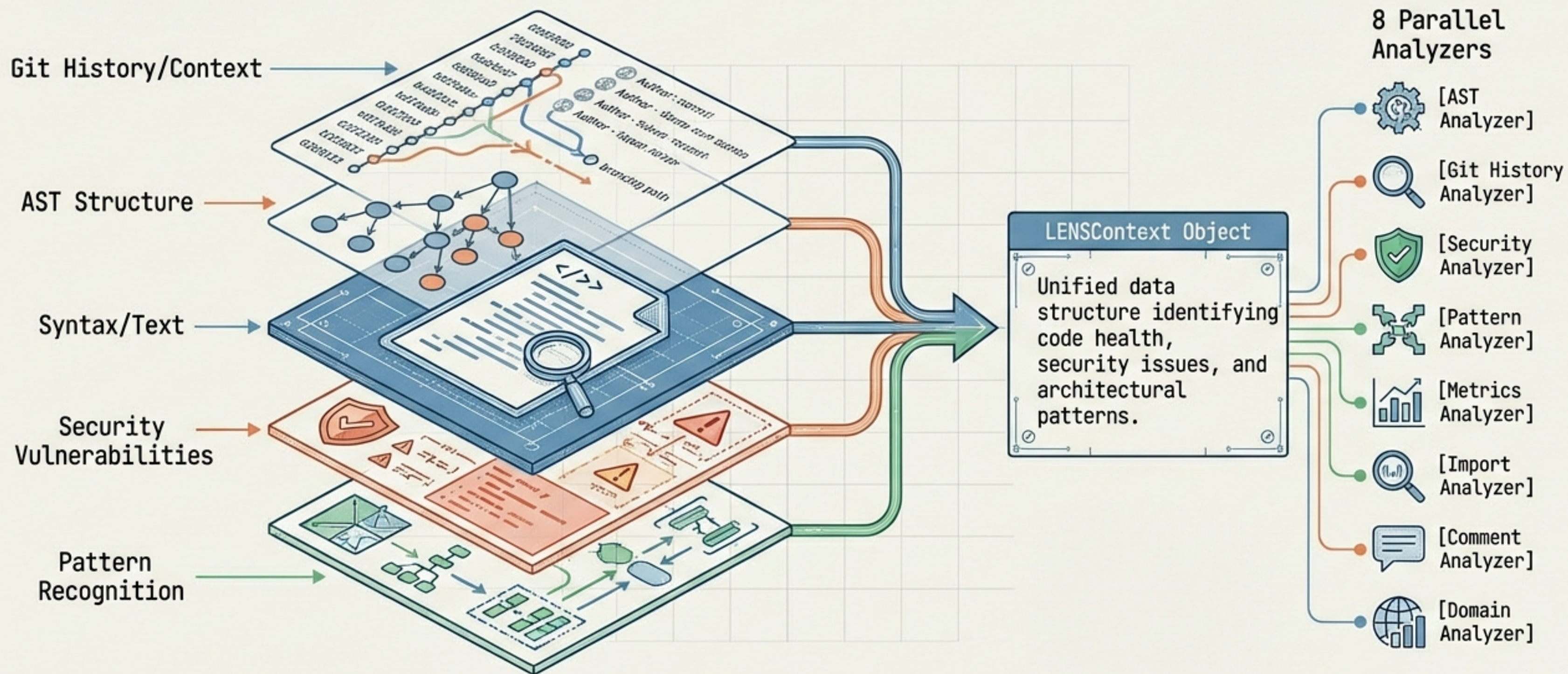
Custom MCP tools & plugins

Zero core changes



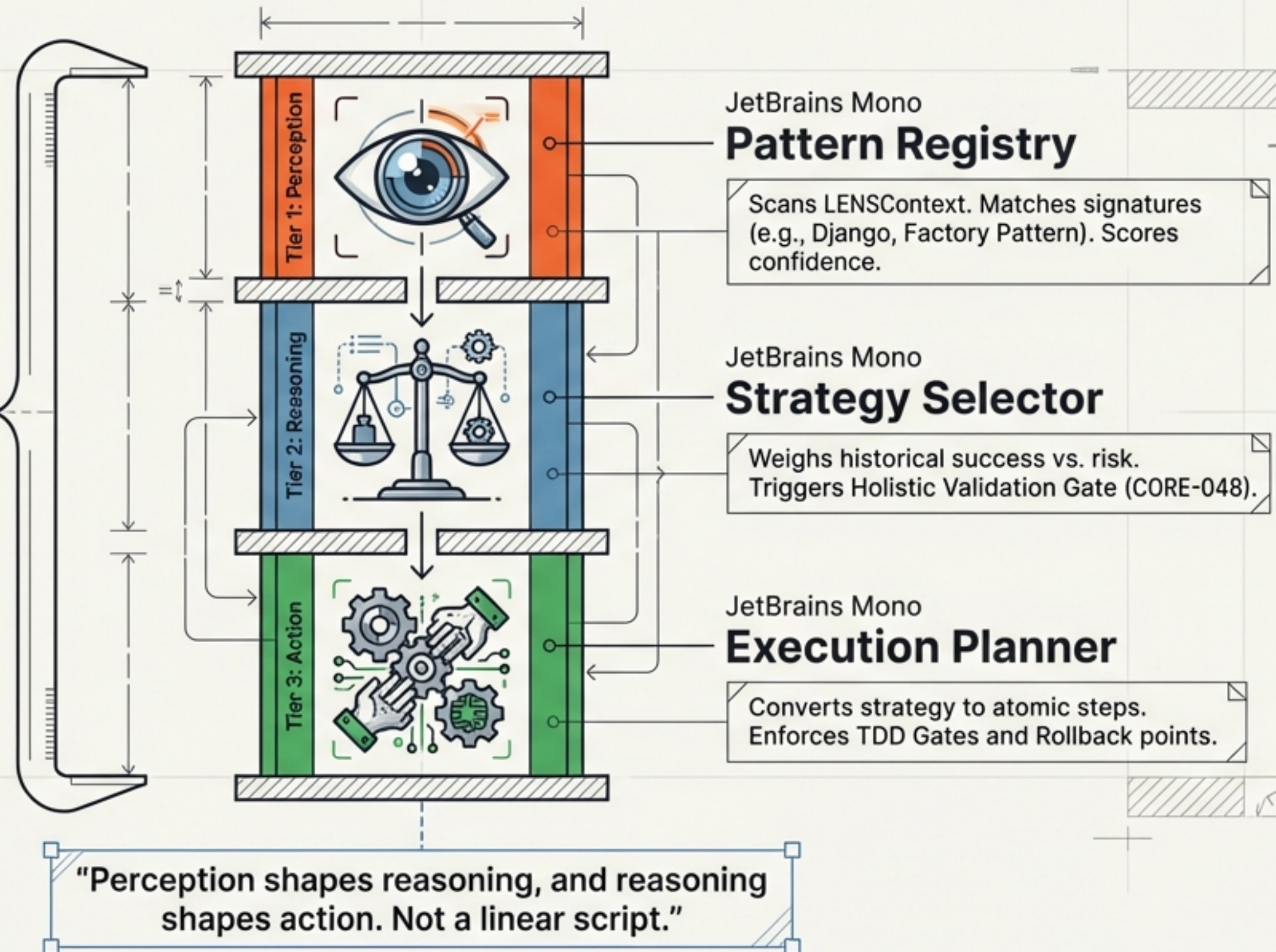
The Senses: LENS Intelligence

Language • Examination • Navigation • Synthesis

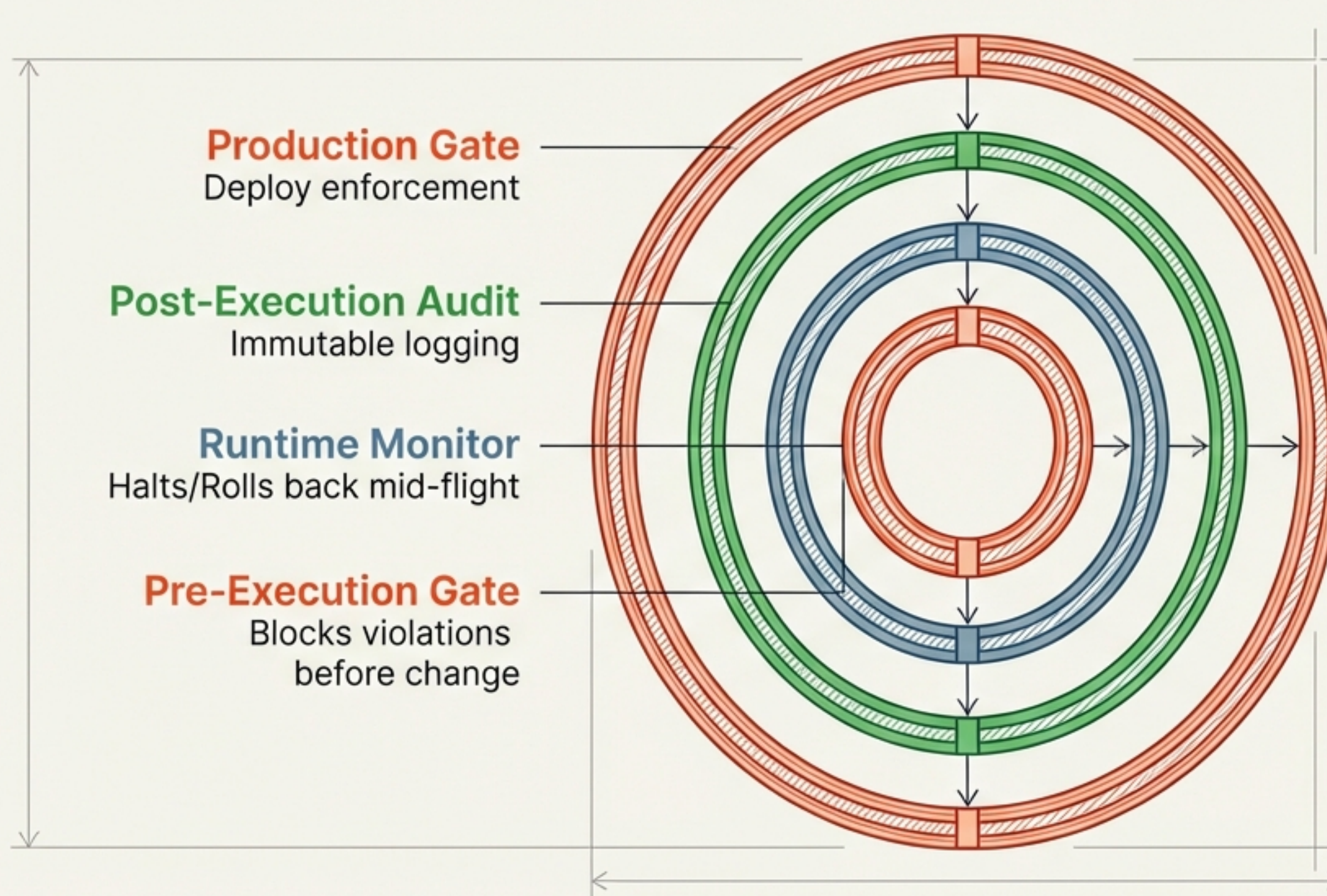


The Brain: A Three-Tier Cognitive Core

Helvetica Now Display
Cognitive Pipeline



The Conscience: Governance Is Not Optional



Sample CORE Rules

- ✓ CORE-008: Mandatory TDD
(No code before failing test)
- ✓ CORE-002: No report sprawl
(Inline results only)
- ✓ CORE-049: MCP-First
Architecture

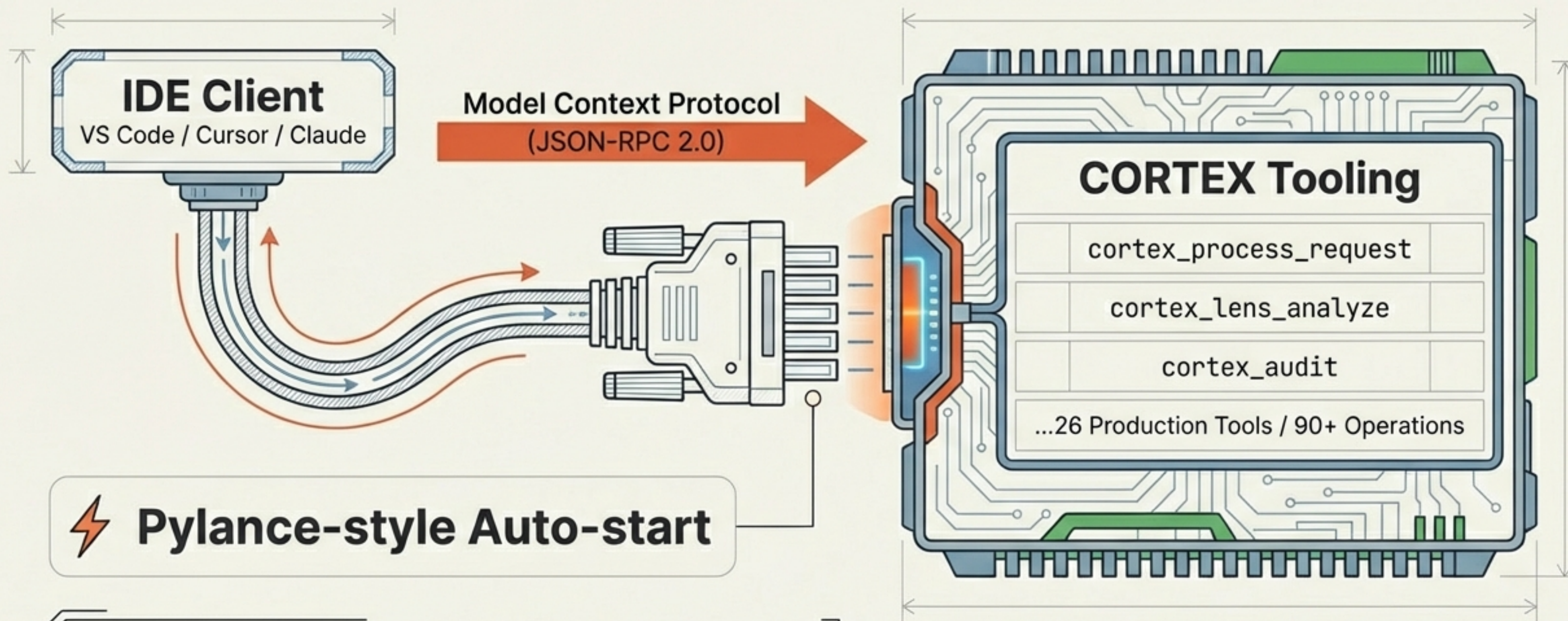
59 CORE Rules
enforced automatically

The Nervous System: Intelligent Orchestration



Work is dispatched to 20+ specialized engines, not generic prompts.

The Hands: MCP-First Architecture



Pylance-style Auto-start



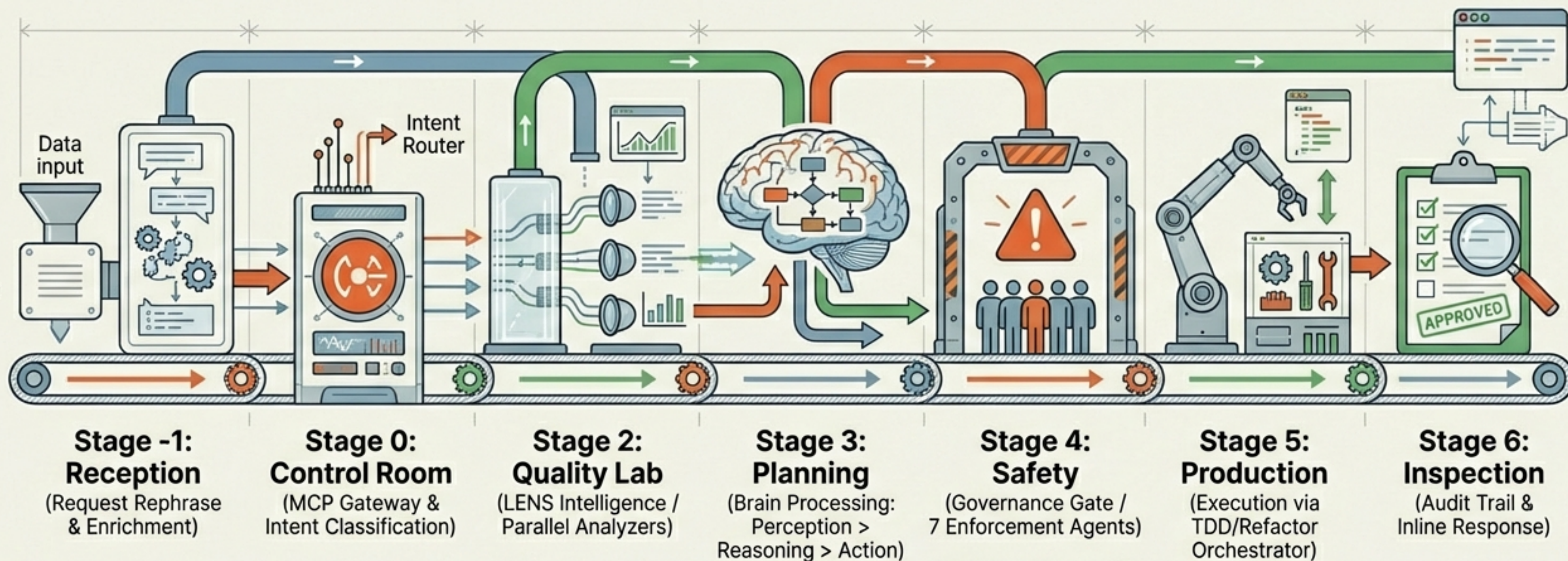
CORE-049: Direct file operations blocked.
All changes must occur via defined API tools.



CORTEX

NotebookLM

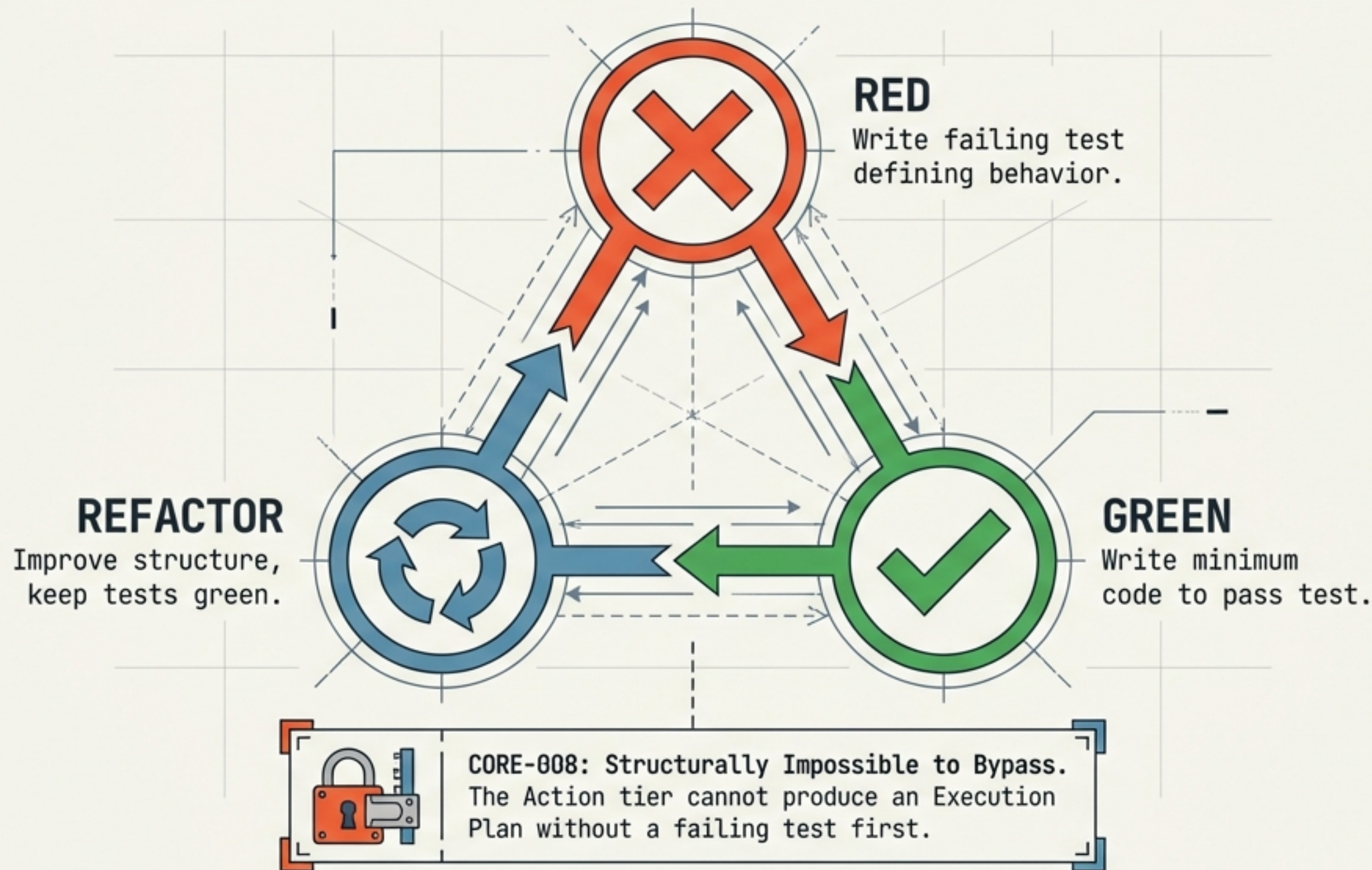
The Workflow: From Intent to Commit



End-to-end Latency: ~300ms for query / 2-8s for implementation.

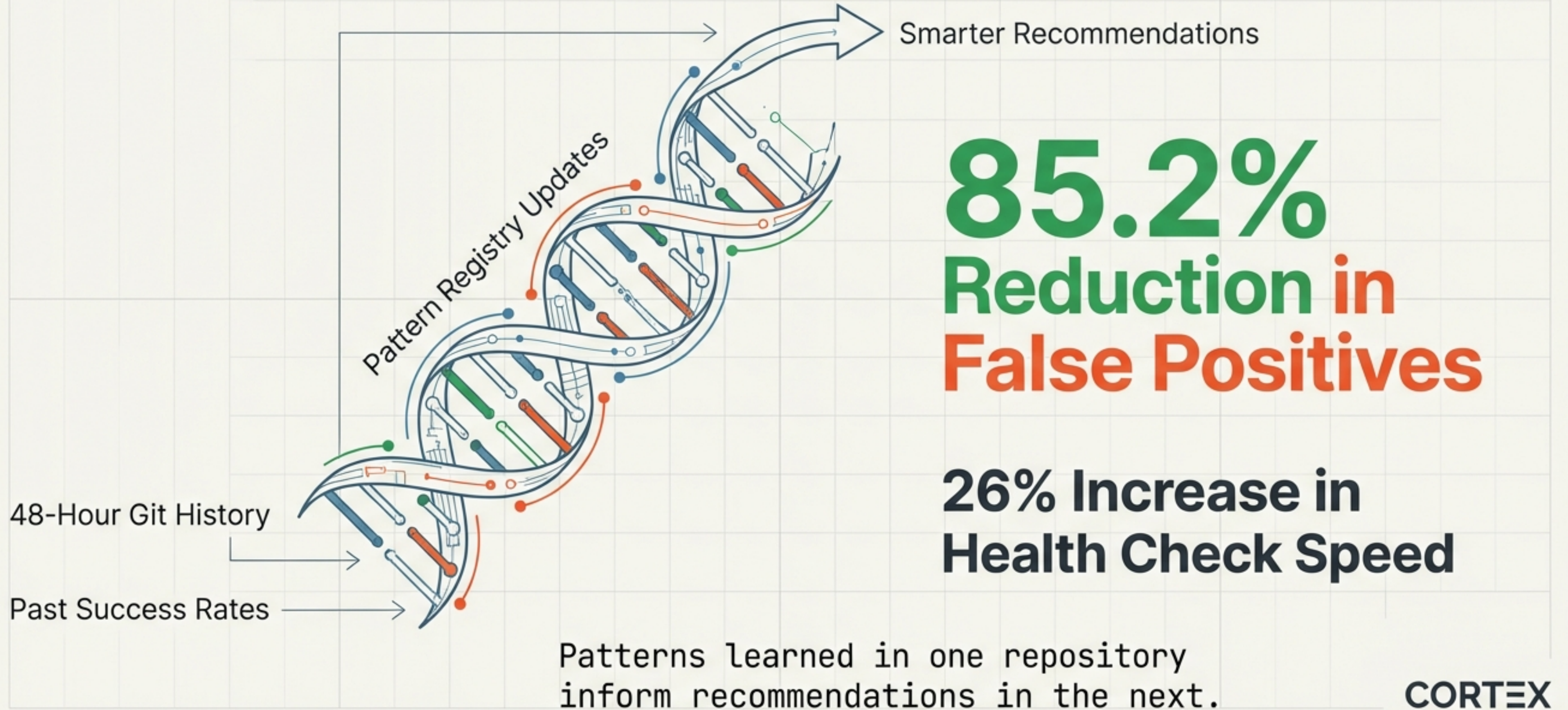
CORE-049: Direct file operations blocked.

Enforced Discipline: The TDD Cycle

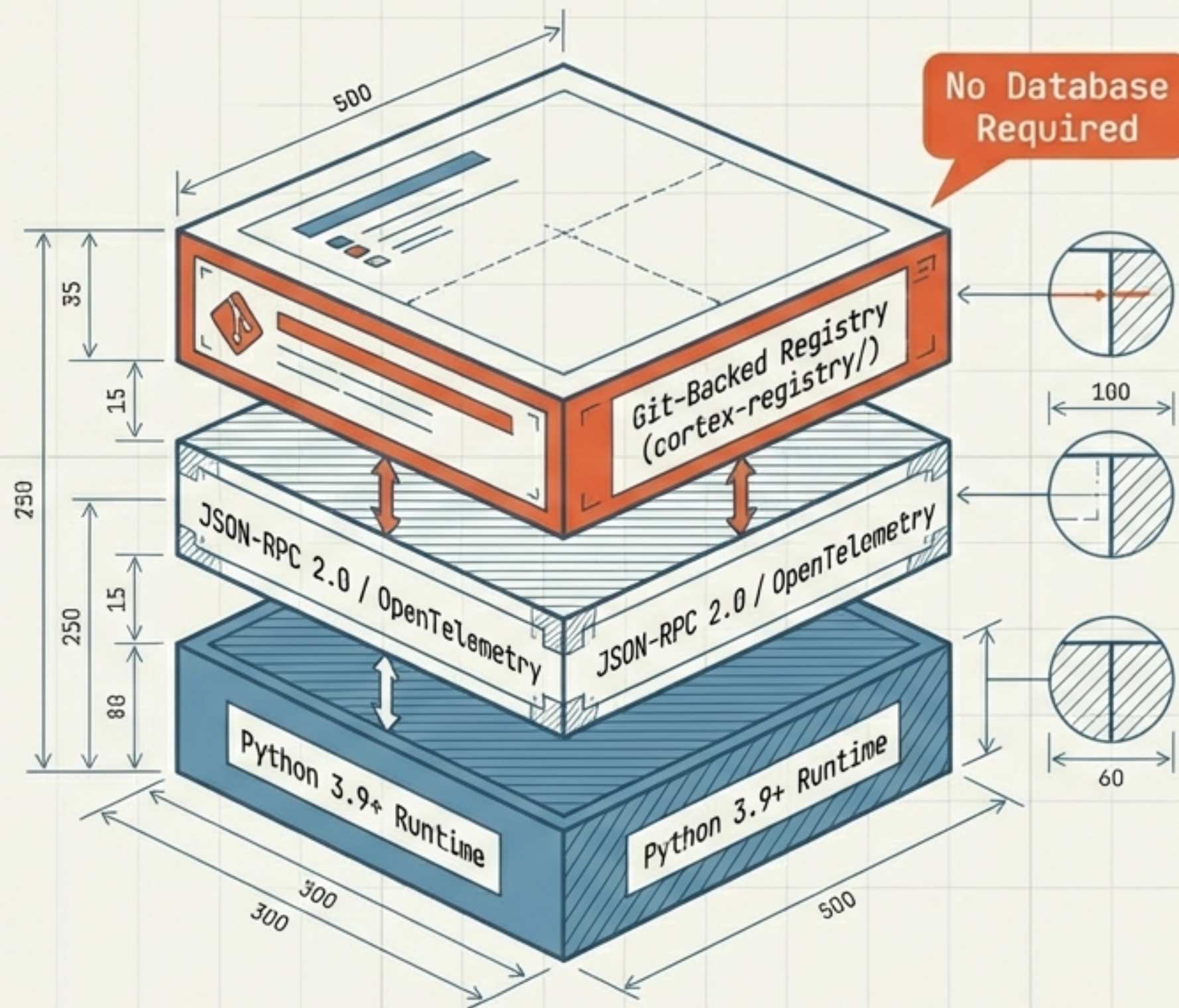


Developers see quality. Leaders see stability.

Memory & Learning: Reducing False Positives



Infrastructure: Configuration as Code



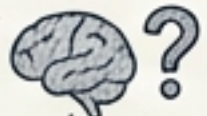
- Human-readable YAML configuration
- Versioned and auditable history
- Rollback via ``git revert``
- ``__wiring_contract__.yaml`` defines orchestrator registration

The Operational Experience

Before CORTEX



Manual checklists



Write code, hope tests follow



Risky refactors



File sprawl (.md reports)

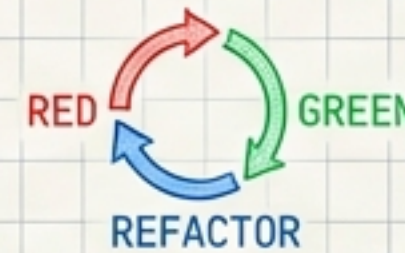
 CORTEX
Integration



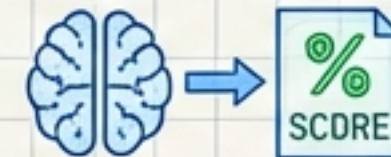
After CORTEX



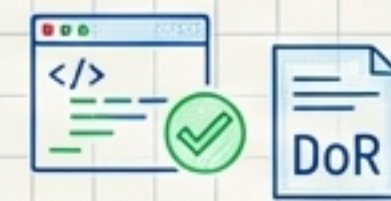
Automated 8-analyzer reports



Enforced RED-GREEN-REFACTOR



Semantic refactoring
with regression scoring

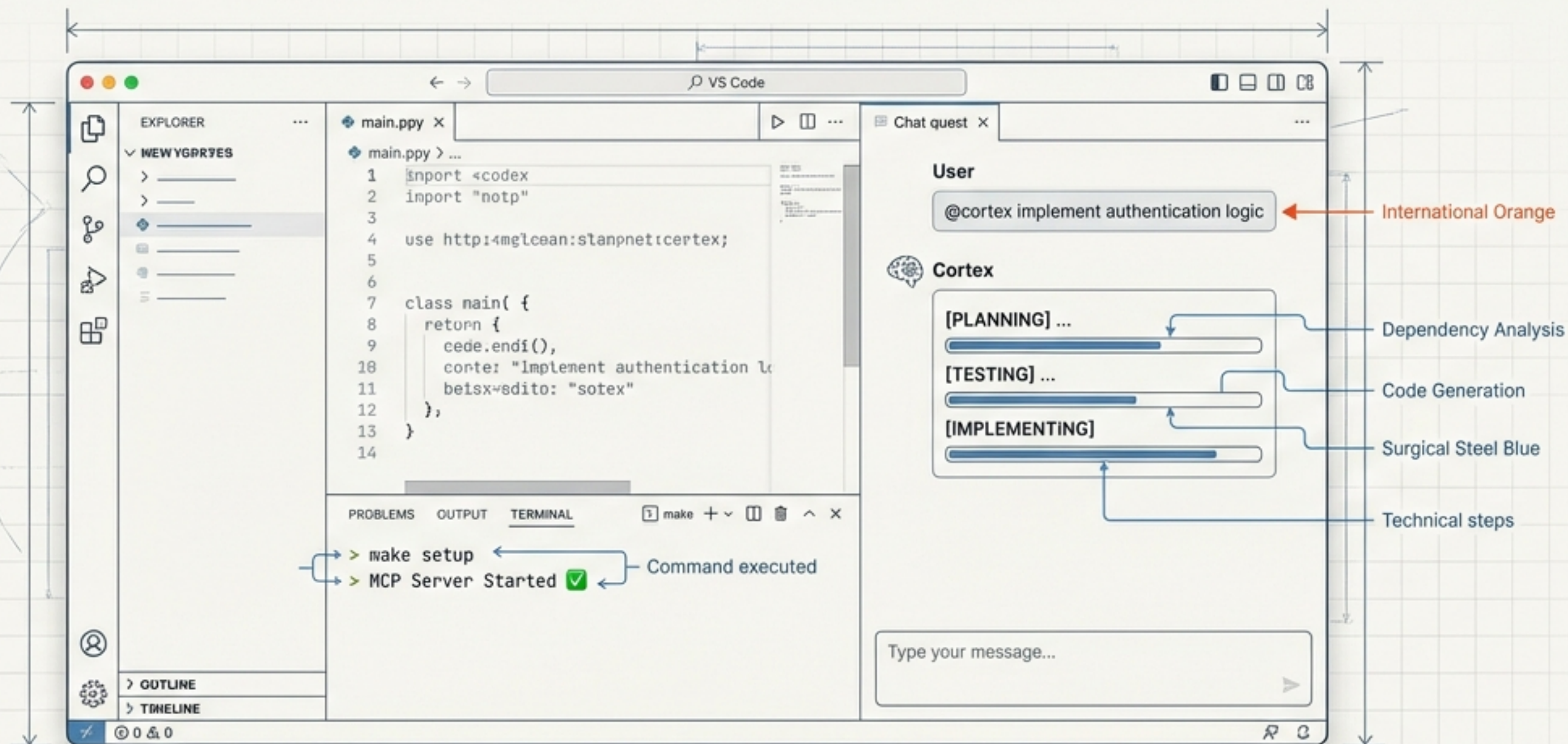


Inline audits &
Definition of Ready (DoR)

User Interaction: Inline responses only. No context switching.

 CORTEX

Getting Started: First Request in 5 Minutes



1. Clone Workspace. 

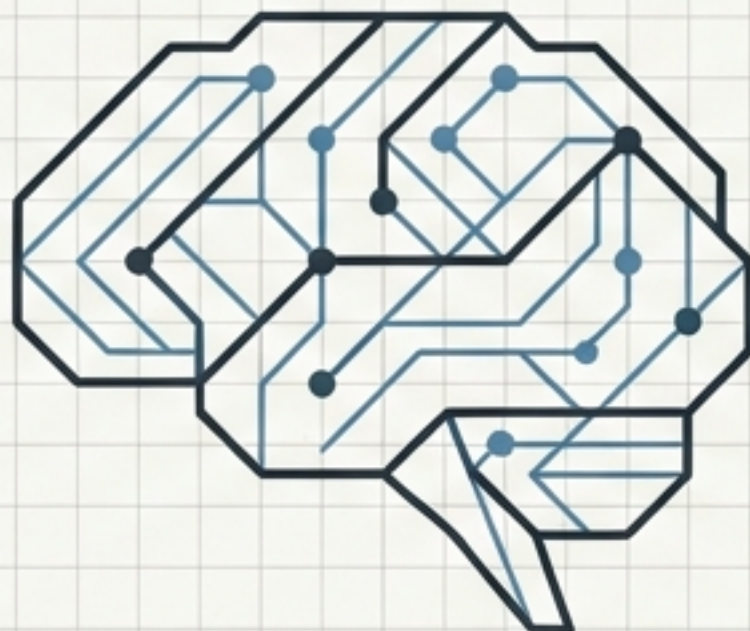
2. Auto-Start (VS Code Pylance-style detection). 

3. Request Feature. 

Simple function implementation in ~1.1 seconds.

Ship Faster. With Confidence.

Governance is architectural. Intelligence is learned. Quality is enforced.



Review the source of truth:
`cortex/__wiring_contract__.yaml`